What if we have cycles?! LAO* Search

- LAO* generalizes AO* for AND/OR graphs with cycles[8]:
  - It uses the same **Expansion Stage** of AO*;
  - But it uses a **Policy Evaluation** (i.e., Policy Iteration or Value Iteration) for the **Cost Revision Stage**;

[8] *Hansen and Zilberstein, Heuristic Search in Cyclic AND/OR Graphs. AAAI, 1998.*

Take-Away Message

- *Heuristic Search* **tends to not explore** the entire state-space to find a solution (plan);
- We need **admissible heuristics** for **optimal** planning;
- The use of **non-admissible heuristics** may be **faster** to find a solution, but such a solution may **not be optimal**;

Why AI Planning is so interesting?!

- The most effective *Planning* algorithms **DO NOT** rely on data or use *Learning*;
- *Symbolic Planning* is **MUCH MORE** effective than *Learning* techniques for several problems:
  - Atari Games[9];

[9] Lipovetzky, Ramírez, and Geffner, *Classical Planning with Simulators: Results on the Atari Video Games*, IJCAI, 2015.

Thank you!

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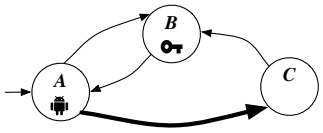
SAPIENZA Università di Roma – MSc. in Engineering in Computer Science

Artificial Intelligence & Machine Learning – June 23, 2022

(Time to complete the test: 2:30 hours)

Part 1 - Artificial Intelligence *(Time to complete Part 1: 100 minutes)*

Consider the following scenario. A robot, *self*, can move across a network of locations shown in the picture below.



To traverse some edges of the network *self* needs a key (specifically the edge in **bold** in the picture).

Non-fluents predicates: $Location(x)$ denoting locations, in our case A, B, C; $Obj(x)$ denoting the objects, in our case only the key key; $connected(x, y)$ where x and y are locations, denoting the edges not requiring a key in the network (see picture above); $connectedKey(x, y)$ where x and y are locations, denoting the edges that need the key to be traversed (see the picture above).

Fluents: $SelfAt(x)$ where x is a location, denoting that *self* is at location x; $At(x, y)$ where x is an object and y a location, denoting that the object x is at location y; $SelfHas(x)$ where x is an object, and denotes that *self* has object x with himself/herself.

Actions as available to *self*:

- $goto(\ell_1, \ell_2)$ to move from location ℓ_1 to location ℓ_2. To do so *self* needs to be in location ℓ_1, moreover if traversing the edge in the network needs the key, *self* need to have the key. The effect of $goto(\ell_1, \ell_2)$ is that *the* new location of *self*, together with all the objects that *self* has with him/her (i.e., the key), will be ℓ_2.
- $pickup(obj)$ to pick up object obj (i.e., the key). To do so *self* needs to be in the same location as obj. The effect of $pickup(obj)$ is that *self* will have obj with himself/herself.
- ~~$drop(obj)$ to drop obj (i.e., the key). To do so *self* needs to have obj with himself/herself. The effect of $drop(obj)$ is that *self* will not have obj with himself/herself and the location of obj will be that of *self*.~~

Initially: *self* is at location A, the key is at location B, and *self* does not have any object with himself/herself. (The non-fluent predicates have their extension as above.)

Exercise 1. Formalize the above scenario as a Basic Action Theory in Reiter's Situation Calculus. Then check by regression:

1. Whether the action $goto(A, C)$ is executable in S_0;
2. Whether the sequence of actions $goto(A, B); pickup(key); goto(B, A); goto(A, C)$ is executable in S_0;
3. Whether, after executing the sequence of actions $goto(A, B); pickup(key); goto(B, A); goto(A, C)$ in S_0, *self* has the key with himself/herself.

Exercise 2. Considering as goal $SelfAt(C)$, formalize the]above scenario as a PDDL domain file and PDDL problem file. Then:

1. Draw the corresponding transition system;
2. Solve planning for achieving the above goal by using backward fixpoint computation, reporting the various stages of the fixpoint computation;
3. Solve planning for achieving the above goal by using forward depth-first search (uniformed), reporting the various steps of the forward search computation.

Exercise 3. Check, by using tableaux, whether the following FOL formula is indeed valid, and if not show a counterexample: $(\,(\forall x.(A(x) \supset B(x))\,) \;\equiv\; (\,(\forall y.A(y)) \supset (\forall z.B(z))\,)$.

Exercise 4. Describe the DPLL procedure for satisfiability and use it to find a model (a satisfying interpretation) of the following set of clauses: $\{\ \{A, B, \neg C\}, \{\neg A, B, C\}, \{\neg A\}, \{\neg B, C\}\ \}$.

1 Exercise 1: solution

Preliminary discussion. Let's write more rigorously the preconditions and effects described above.

- $goto(\ell_1, \ell_2)$
  - PRE: $SelfAt(\ell_1) \wedge (connected(\ell_1, \ell_2) \ \vee \ connectedKey(\ell_1, \ell_2) \wedge SelfHas(key))$
  - EFF: $(SelfAt(\ell_2) \wedge \neg SelfAt(\ell_1)) \wedge \forall x.PRE[SelfHas(x)] \supset At(x, \ell_2) \wedge \neg At(x, \ell_1)$
- $pickup(obj)$
  - PRE: $\exists \ell.SelfAt(\ell) \wedge At(obj, \ell)$
  - EFF: $SelfHas(obj)$

Basic Action Theory in Reiter's Situation Calculus. Now we formalize the scenario as a Basic Action Theory in Reiter's Situation Calculus.

- **Precondition Axioms.** They can be obtained immediately after the preliminary consideration above:

$$\begin{aligned} Poss(goto(\ell_1, \ell_2), s) &\equiv SelfAt(\ell_1, s) \wedge (connected(\ell_1, \ell_2) \ \vee \ connectedKey(\ell_1, \ell_2) \wedge SelfHas(key, s)) \\ Poss(pickup(obj), s) &\equiv \exists \ell.SelfAt(\ell, s) \wedge At(obj, \ell, s) \end{aligned}$$

- **Successor State Axioms.** To obtain them, we first write the effects using situations:

$$\begin{aligned} & a = goto(\ell_1, \ell_2) \supset \\ & \quad (((SelfAt(\ell_2, do(a, s)) \wedge \neg SelfAt(\ell_1, do(a, s))) \wedge \\ & \quad \forall x.(SelfHas(x, s) \supset At(x, \ell_2, do(a, s)) \wedge \neg At(x, \ell_1, do(a, s))))) \\ & a = pickup(obj) \supset SelfHas(obj, do(a, s)) \end{aligned}$$

  Then we put them in normal form with only one fluent on the right-hand side of the implication. We can apply the following equivalence $[\forall \vec{x}.(\varphi_1(\vec{x}) \supset (\forall \vec{y}.\varphi_2(\vec{x}, \vec{y}) \supset \varphi_3(\vec{x}, \vec{y})))] \equiv [\forall \vec{x}, \vec{y}.(\varphi_1(\vec{x}) \wedge \varphi_2(\vec{x}, \vec{y}) \supset \varphi_3(\vec{x}, \vec{y}))]$:

$$\begin{aligned} & a = goto(\ell_1, \ell_2) \supset SelfAt(\ell_2, do(a, s)) \\ & a = goto(\ell_1, \ell_2) \supset \neg SelfAt(\ell_1, do(a, s))) \\ & a = goto(\ell_1, \ell_2) \wedge SelfHas(x, s) \supset At(x, \ell_2, do(a, s)) \\ & a = goto(\ell_1, \ell_2) \wedge SelfHas(x, s) \supset \neg At(x, \ell_1, do(a, s)) \\ & a = pickup(obj) \supset SelfHas(obj, do(a, s)) \end{aligned}$$

  Note that all free variables in the axioms above are universally quantified. But if a variable appears only on the left-hand side of the implication we can apply the following equivalence $[\forall \vec{x}, \vec{y}.\varphi_1(\vec{x}, \vec{y}) \supset \varphi_2(\vec{x})] \equiv [\forall \vec{x}.(\exists \vec{y}.\varphi_1(\vec{x}, \vec{y})) \supset \varphi_2(\vec{x})]$. Hence we get:

$$\begin{aligned} & (\exists \ell_1.a = goto(\ell_1, \ell_2)) \supset SelfAt(\ell_2, do(a, s)) \\ & (\exists \ell_2.a = goto(\ell_1, \ell_2)) \supset \neg SelfAt(\ell_1, do(a, s))) \\ & (\exists \ell_1.a = goto(\ell_1, \ell_2)) \wedge SelfHas(x, s) \supset At(x, \ell_2, do(a, s)) \\ & (\exists \ell_2.a = goto(\ell_1, \ell_2) \wedge SelfHas(x, s)) \supset \neg At(x, \ell_1, do(a, s)) \\ & a = pickup(obj) \supset SelfHas(obj, do(a, s)) \end{aligned}$$

  Now by applying "*explanation closure*" we finally get the **successor state axioms**:

$$\begin{aligned} SelfAt(\ell, do(a, s)) &\equiv \exists \ell_1.a = goto(\ell_1, \ell) \ \vee \ SelfAt(\ell, s) \wedge \neg \exists \ell_2.a = goto(\ell, \ell_2) \\ At(x, \ell, do(a, s)) &\equiv \exists \ell_1.a = goto(\ell_1, \ell) \wedge SelfHas(x, s) \ \vee \ At(x, \ell, s) \wedge \neg(\exists \ell_2.a = goto(\ell, \ell_2) \wedge SelfHas(x, s)) \\ SelfHas(obj, do(a, s)) &\equiv a = pickup(obj) \ \vee \ SelfHas(obj, s) \end{aligned}$$

- **Initial situation description.** Remember all free variables universally quantified:

  $connected(\ell_1, \ell_2) \equiv (\ell_1 = A \wedge \ell_2 = B) \vee (\ell_1 = B \wedge \ell_2 = A) \vee (\ell_1 = C \wedge \ell_2 = B))$ –it is not a fluent

  $connectedKey(\ell_1, ell_2)) \equiv (\ell_1 = A \wedge \ell_2 = C)$ –it is not a fluent

  $SelfAt(\ell, S_0) \equiv (\ell = A)$

  $At(key, \ell, S_0) \equiv (\ell = B)$

  $\neg \exists x.SelfHas(x, S_0)$

1. Check by regression whether the action $goto(A, C)$ is executable in S_0

$$\mathcal{R}[Poss(goto(A, C), S_0)] = SelfAt(A, S_0) \wedge (connected(A, C) \vee connectedKey(A, C) \wedge SelfHas(key, S_0))$$

Considering the initial situation description, we get that $connected(A, C)$ is false and while $connectedKey(A, C)$ is true, it is not the case that $SelfHas(key, S_0)$. So $goto(A, C)$ is not executable in S_0.

2. Check by regression whether the sequence of actions $goto(A, B); pickup(key); goto(B, A); goto(A, C)$ is executable in S_0
We have to check each of the following:

$$\begin{array}{ll}
Poss(goto(A, B), S_0) & \\
Poss(pickup(key), S_1) & S_1 = do(goto(A, B), S_0) \\
Poss(goto(B, A), S_2) & S_2 = do(pickup(key), S_1) = do(pickup(key), do(goto(A, B), S_0)) \\
Poss(goto(A, C), S_3) & S_3 = do(goto(B, A), S_2) = \\
 & \quad do(goto(B, A), do(pickup(key), S_1)) = do(goto(B, A), do(pickup(key), do(goto(A, B), S_0)))
\end{array}$$

We check each of them by regression:

$$\mathcal{R}[Poss(goto(A, B), S_0)] = SelfAt(A, S_0) \wedge (connected(A, B) \vee connectedKey(A, B) \wedge SelfHas(key, S_0))$$

Considering the initial situation description this is true since $SelfAt(A, S_0)$ and $connected(A, B)$ both hold in the initial situation.

$$\begin{array}{rcl}
\mathcal{R}[Poss(pickup(key), S_1)] & = & \exists \ell.\mathcal{R}[SelfAt(\ell, S_1)] \wedge \mathcal{R}[At(key, \ell, S_1)] \\
 & = & \exists \ell.\mathcal{R}[\exists \ell_1.goto(A, B) = goto(\ell_1, \ell) \vee \\
 & & \quad SelfAt(\ell, s) \wedge \neg \exists \ell_2.goto(A, B) = goto(\ell, \ell_2)] \wedge \mathcal{R}[At(key, \ell, do(goto(A, B), S_0))] \\
 & = & true \wedge \mathcal{R}[At(key, B, do(goto(A, B), S_0))] \\
\mathcal{R}[At(key, B, do(goto(A, B), S_0))] & = & \exists \ell_1.goto(A, B) = goto(\ell_1, \ell) \wedge SelfHas(key, S_0) \vee \\
 & & \quad At(key, \ell, S_0) \wedge \neg(\exists \ell_2.goto(A, B) = goto(\ell, \ell_2) \wedge SelfHas(key, S_0)) \\
 & = & goto(A, B) = goto(A, B) \wedge SelfHas(key, S_0) \vee \\
 & & \quad At(key, B, S_0) \wedge \neg(\exists \ell_2.goto(A, B) = goto(B, \ell_2) \wedge SelfHas(key, S_0))] \\
 & = & SelfHas(key, S_0) \vee At(key, B, S_0) \\
 & = & true
\end{array}$$

So $pickup(key)$ is indeed executable after $goto(A, B)$.

$$\begin{array}{rcl}
\mathcal{R}[Poss(goto(B, A), S_2] & = & \mathcal{R}[SelfAt(B, S_2)] \wedge (connected(B, A) \vee connectedKey(B, A) \wedge \mathcal{R}[SelfHas(key, S_2)]) \\
 & = & \mathcal{R}[SelfAt(B, S_2)] \wedge true \\
\mathcal{R}[SelfAt(B, do(pickup(key), S_1))] & = & \exists \ell_1.pickup(key) = goto(\ell_1, B) \vee \mathcal{R}[SelfAt(B, S_1)] \wedge \neg \exists \ell_2.pickup(key) = goto(B, \ell_2) \\
 & = & false \vee \mathcal{R}[SelfAt(B, S_1)] \\
\mathcal{R}[SelfAt(B, do(goto(A, B), S_0))] & = & \exists \ell_1.goto(A, B) = goto(\ell_1, B) \vee SelfAt(B, S_0)) \wedge \neg \exists \ell_2.goto(A, B) = goto(B, \ell_2) \\
 & = & true
\end{array}$$

So $goto(B, A)$ is indeed executable after $goto(A, B)$ followed by $pickup(key)$.

$$\begin{array}{rcl}
\mathcal{R}[Poss(goto(A, C), S_3) & = & \mathcal{R}[SelfAt(A, S_3)] \wedge (connected(A, C) \vee connectedKey(A, C) \wedge \mathcal{R}[SelfHas(key, S_3)]) \\
 & = & (true \wedge (false \vee true \wedge \mathcal{R}[SelfHas(key, S_3)])] \\
\mathcal{R}[SelfHas(key, do(goto(B, A), S_2))])] & = & goto(B, A) = pickup(key)) \vee \mathcal{R}[SelfHas(key, S_2)] \\
 & = & false \vee \mathcal{R}[SelfHas(key, S_2)] \\
\mathcal{R}[SelfHas(key, do(pickup(key), S_1))] & = & pickup(key) = pickup(key) \vee \mathcal{R}[SelfHas(key, S_1)] \\
 & = & true
\end{array}$$

So $goto(A, C)$ is indeed executable after $goto(A, B)$ followed by $pickup(key)$, followed by $goto(B, A)$. So we can conclude that the sequence $goto(A, B); pickup(key); goto(B, A); goto(A, C)$ is executable in S_0

3. Check by regression whether, after executing the sequence of actions $goto(A, B); pickup(key); goto(B, A); goto(A, C)$ in S_0, *self* has the key with himself/herself. As before we use $S_1 = do(goto(A, B), S_0)$, $S_2 = do(pickup(key), S_1)$, $S_3 = do(goto(B, A), S_2)$ and $S_4 = do(goto(A, C), S_3)$.

$$\begin{array}{rcl}
\mathcal{R}[SelfHas(key, do(goto(A, C), S_3))] & = & goto(A, C) = pickup(key) \vee \mathcal{R}[SelfHas(key, S_3)] \\
 & = & false \vee \mathcal{R}[SelfHas(key, S_3)] \\
\mathcal{R}[SelfHas(key, do(goto(B, A), S_2))] & = & goto(B, A) = pickup(key) \vee \mathcal{R}[SelfHas(key, S_2)] \\
 & = & false \vee \mathcal{R}[SelfHas(key, S_2)] \\
\mathcal{R}[SelfHas(key, do(pickup(key), S_1))] & = & pickup(key) = pickup(key) \vee \mathcal{R}[SelfHas(key, S_1)] \\
 & = & true
\end{array}$$

Hence the projection above is indeed correct.

2 Exercise 2: solution

PDDL

- **Domain file:**

```
(define (domain ex2dom)
(:requirements :adl)

(:types
    object
    loc
)

(:constants
     key - object
)

(:predicates
    (connected ?l1 ?l2 - loc)      ;; nonfluent
    (connectedKey ?l1 ?l2 - loc)   ;; nonfluent
    (SelfAt ?l - loc)              ;; fluent
    (At ?o - object ?l - loc)      ;; fluent
    (SelfHas ?o - object)          ;; fluent
)


(:action goto
:parameters (?l1 ?l2 - loc)
:precondition
        (and (SelfAt ?l1) (or (connected ?l1 ?l2)
                              (and (connectedKey ?l1 ?l2) (SelfHas key))))
:effect
        (and (SelfAt ?l2)
             (not (SelfAt ?l1))
             (forall (?o - object)
               (when (SelfHas ?o) (and (At ?o ?l2)
                                       (not (At ?o ?l1)))
               )
             )
        )
)

(:action pickup
:parameters (?o - object)
:precondition (exists (?l - loc) (and (SelfAt ?l) (At ?o ?l)))
:effect  (SelfHas ?o)
)
```

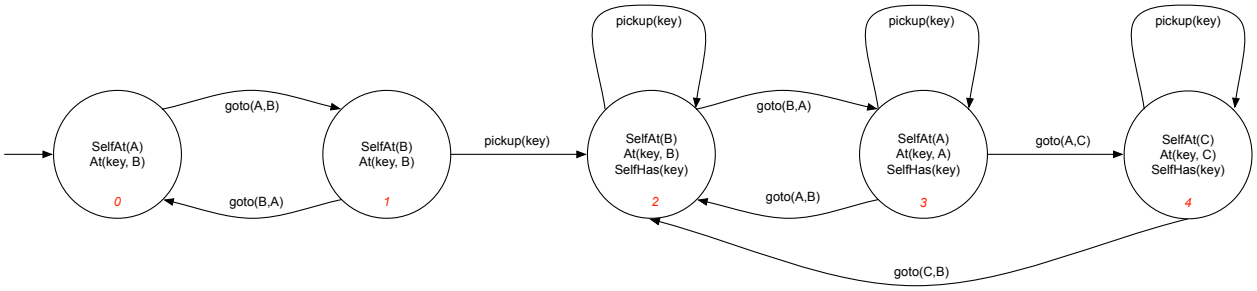
- **Problem file:**

```
(define (problem ex2prob)
(:domain ex2dom)
(:objects
        key - object
        A - loc
        B - loc
        C - loc
)
```

```
    (:init
          (connected A B)
          (connected B A)
          (connectedKey A C)
          (connected C B)
          (SelfAt A)
          (At key B)
          ;; SelfHas is false for every object
    )

    (:goal
          (SelfAt C)
    )
    )
```

Corresponding transition system. Here is the transition system corresponding to the planning domain + initial state, considering that the only object present is the key (note every fluent atom not reported in the diagram is false):



Planning by backward fixpoint. We compute the plan by backward fixpoint:

$$
\begin{array}{lcll}
Win_0 & = & [[G]] & = \{4\} \\
Win_1 & = & Win_0 \cup PreE(Win_0) & = \{3,4\} \\
Win_2 & = & Win_1 \cup PreE(Win_1) & = \{2,3,4\} \\
Win_3 & = & Win_2 \cup PreE(Win_2) & = \{1,2,3,4\} \\
Win_4 & = & Win_3 \cup PreE(Win_3) & = \{0,1,2,3,4\} \\
Win_5 & = & Win_4 \cup PreE(Win_4) & = \{0,1,2,3,4\} \quad \textit{fixpoint found!}
\end{array}
$$

Since the initial state 0 is in the fixpoint $Win = \bigcup_i Win_i$ we know that a plan exists, and we can actually compute a (universal) plan as follows:

$$
\begin{array}{lcll}
\omega(0) & = & goto(A,B) & -- \quad 0 \in Win_4,\ \delta(0, goto(A,B)) \in Win_3 \\
\omega(1) & = & pickup(key) & -- \quad 1 \in Win_3,\ \delta(1, pickup(key)) \in Win_2 \\
\omega(2) & = & goto(B,A) & -- \quad 2 \in Win_2,\ \delta(2, goto(B,A)) \in Win_1 \\
\omega(3) & = & goto(A,C) & -- \quad 3 \in Win_1,\ \delta(3, goto(A,C)) \in Win_0 \\
\omega(4) & = & win! & -- \quad \text{self is in goal state}
\end{array}
$$

Planning by deep-first search. Here is the algorithm:

```
plan DepthFirstSearch () {
  Stack t;
  t.push( (s0,empty) );
  while (!t.empty()) {
    StatePlan (s,p) = t.pop();
    if (!marked(s)) {
      mark(s);
      if (goal(s)) return p;
      forall (a in alpha(s))
         t.push( (delta(s,a), p+a) );
    }
  }
  return noplan;
}
```

We execute the algorithm on our problem (we intelligently perform the pushes in the stack, for simplicity).

```
t.push(0,empty)

(0,empty) <- t.pop()
mark(0)
t.push((1,goto(A,B)))

(1,goto(A,B)) <- t.pop()
mark(1)
t.push((0,goto(A,B);goto(B,A)))
t.push((2,goto(A,B);pickup(key)))

(2,goto(A,B);pickup(key)) <- t.pop()
mark(2)
t.push((2,goto(A,B);pickup(key);pickup(key)))
t.push((3,goto(A,B);pickup(key);goto(B,A)))

(3,goto(A,B);pickup(key);goto(B,A)) <- t.pop()
mark(3)
t.push((2,goto(A,B);pickup(key);goto(B,A);pickup(key)))
t.push((3,goto(A,B);pickup(key);goto(B,A);goto(A,B)))
t.push((4,goto(A,B);pickup(key);goto(B,A);goto(A,C)))

(4,goto(A,B);pickup(key);goto(B,A);goto(A,C)) <- t.pop()
mark(4)
//4 is goal!
return goto(A,B);pickup(key);goto(B,A);goto(A,C)
```

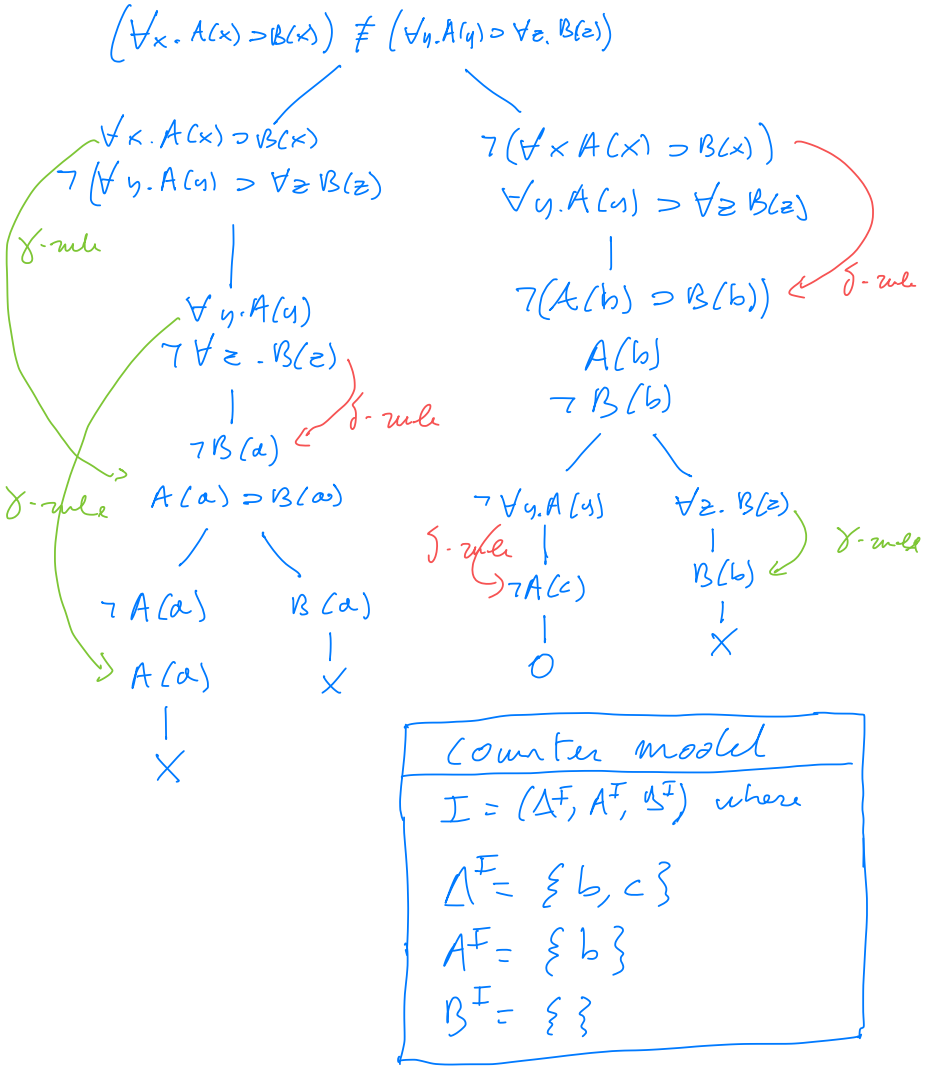
So we return the plan $goto(A, B); pickup(key); goto(B, A); goto(A, C)$, which indeed reaches the goal.

3 Exercise 3: solution

To show that $(\,(\forall x.(A(x) \supset B(x)))\,) \;\equiv\; (\,(\forall y.A(y)) \supset (\forall z.B(z))\,)$ is valid we try to show that

$$(\,(\forall x.(A(x) \supset B(x)))\,) \;\not\equiv\; (\,(\forall y.A(y)) \supset (\forall z.B(z))\,)$$

is unsatisfiable by using tableaux.



The tableau remains open, and a counter example can be build: $I = (\Delta^I, A^I, B^I)$ with $\Delta^I = \{c, b\}, A^I = \{c\}, B^I = \{\}$. I is indeed is a counter model: $I \not\models \forall x.(A(x) \supset B(x))$, since $A^I \not\subseteq B^I$, while $I \models (\forall y.A(y)) \supset (\forall z.B(z))$ since $\forall y.A(y)$ is false and hence the implication itself is trivially true.

4 Exercise 4: solution

We remind the DPLL algorithm based on unit propagation and splitting rule:

```
DPLL(phi, I')
  (psi, I) := UnitPropagation(phi, I');
  if psi contains {} then
      return ({{}}, emptyset)
  elseif psi = {} then
      return ({}, I )
  else
      select a literal lambda in a clause C in psi;  //splitting rule
      if DPLL(psi union {{lambda}}, I) = ({}, I'') then return ({}, I'')
      else return DPLL(psi union {{not lambda}}, I)

UnitPropagation(phi, I)
   while phi contains a unit clause {lambda}
     phi := phi|lambda;
     if lambda = p, then I(p) := true;
     if lambda = not p, then I(p) := false
   end;
   return (phi,I)
```

We apply the DPLL algorithm to our case:

$DPLL(\{\ \{A, B, \neg C\}, \{\neg A, B, C\}, \{\neg A\}, \{\neg B, C\}\ \}, \{\})) \rightarrow$
$UnitPropagation(\{\ \{A, B, \neg C\}, \{\neg A, B, C\}, \{\neg A\}, \{\neg B, C\}\ \}, \{\})) \rightarrow (\{\ \{B, \neg C\}, \{\neg B, C\}\ \}, \{\neg A\})$

$DPLL(\{\ \{B, \neg C\}, \{\neg B, C\}\ \}, \{\neg A\}) \rightarrow$
apply splitting rule, e.g., on B
$UnitPropagation(\{\ \{B, \neg C\}, \{\neg B, C\}, \{B\}\ \}, \{\neg A, B\})) \rightarrow (\{\ \{C\}\ \}, \{\neg A, B\})$

$DPLL(\{\ \{C\}\ \}, \{\neg A, B\}) \rightarrow$
$UnitPropagation((\{\ \{C\}\ \}, \{\neg A, B\}) \rightarrow (\{\}, \{\neg A, B, C\})$

return $(\{\}, \{\neg A, B, C\})$

Hence the formula is satisfiable, and a model for it is $A = false, B = true, C = true$.